Matlab tutorial (Basics)

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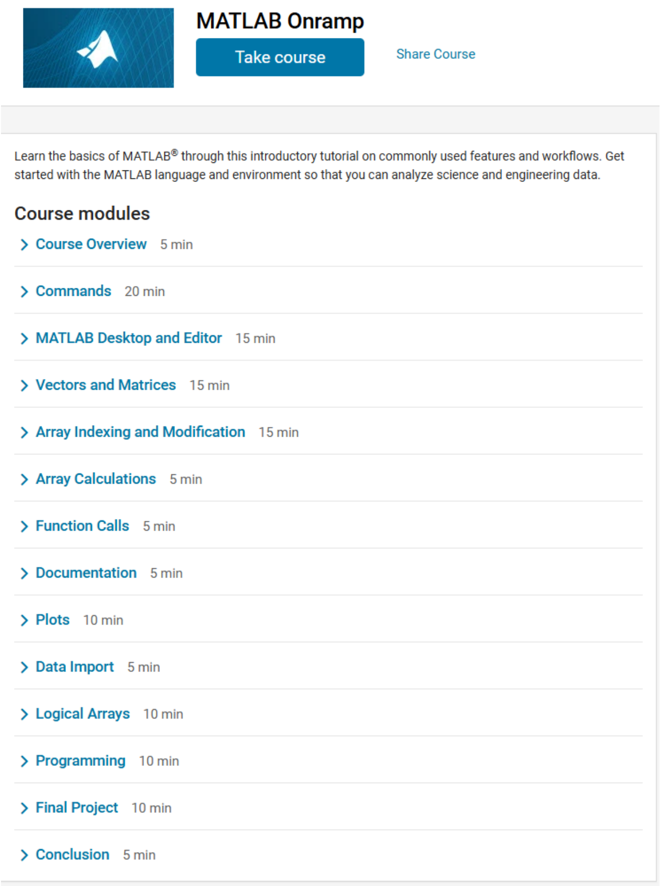
Self-paced Matlab online tutorial

https://matlabacademy.mathworks.com/?page=1&sort=featured



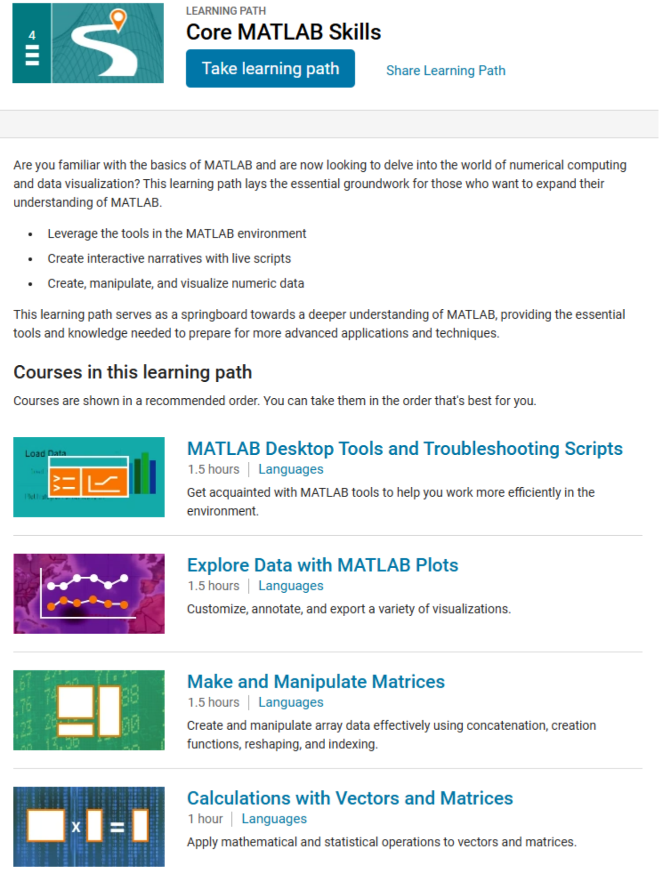
MATLAB Onramp

Take course   Share Course

Learn the basics of MATLAB® through this introductory tutorial on commonly used features and workflows. Get started with the MATLAB language and environment so that you can analyze science and engineering data.

Course modules

- Course Overview 5 min
- Commands 20 min
- MATLAB Desktop and Editor 15 min
- Vectors and Matrices 15 min
- Array Indexing and Modification 15 min
- Array Calculations 5 min
- Function Calls 5 min
- Documentation 5 min
- Plots 10 min
- Data Import 5 min
- Logical Arrays 10 min
- Programming 10 min
- Final Project 10 min
- Conclusion 5 min



LEARNING PATH

Core MATLAB Skills

Take learning path   Share Learning Path

Are you familiar with the basics of MATLAB and are now looking to delve into the world of numerical computing and data visualization? This learning path lays the essential groundwork for those who want to expand their understanding of MATLAB.

- Leverage the tools in the MATLAB environment
- Create interactive narratives with live scripts
- Create, manipulate, and visualize numeric data

This learning path serves as a springboard towards a deeper understanding of MATLAB, providing the essential tools and knowledge needed to prepare for more advanced applications and techniques.

Courses in this learning path

Courses are shown in a recommended order. You can take them in the order that's best for you.

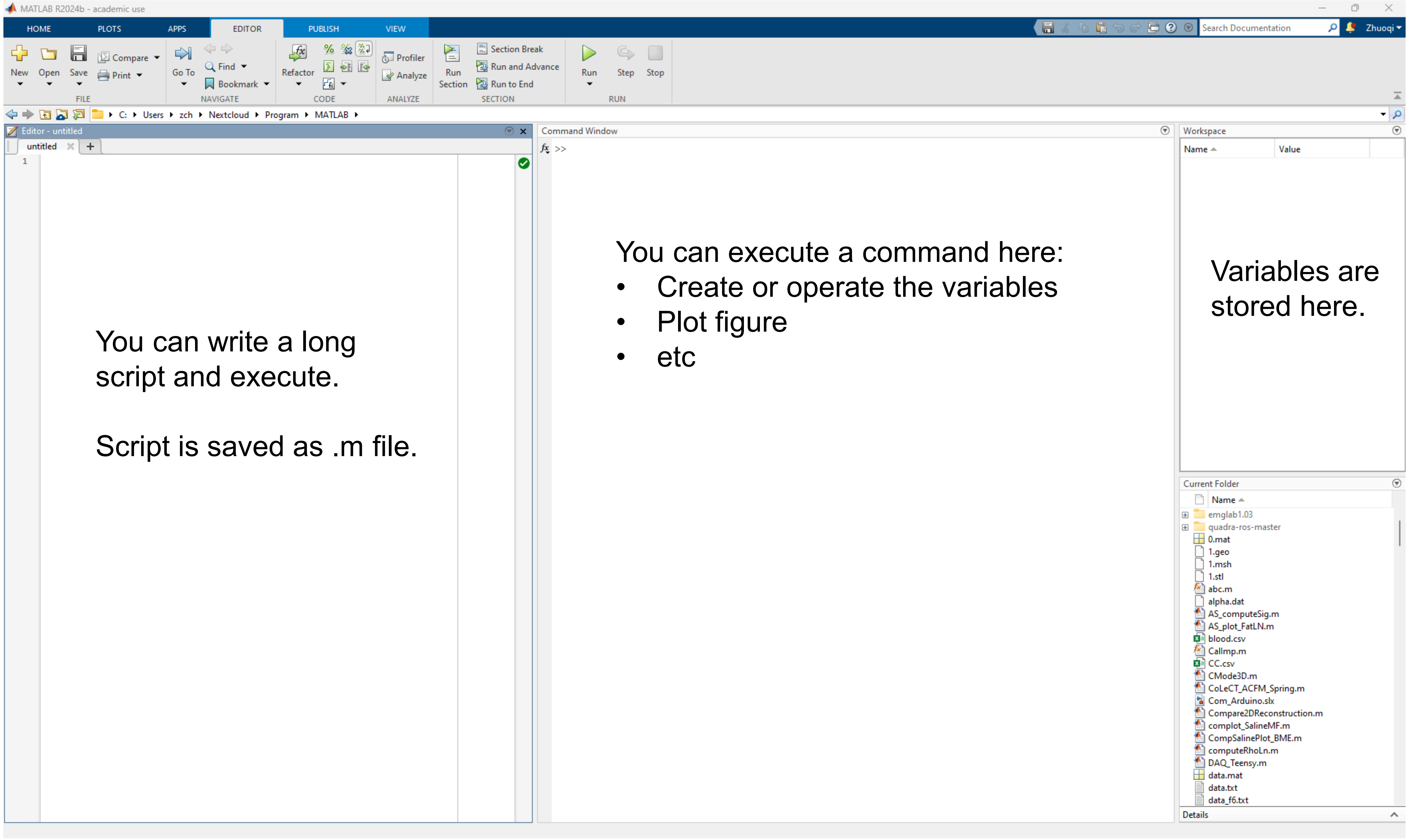
MATLAB Desktop Tools and Troubleshooting Scripts
1.5 hours | Languages
Get acquainted with MATLAB tools to help you work more efficiently in the environment.

Explore Data with MATLAB Plots
1.5 hours | Languages
Customize, annotate, and export a variety of visualizations.

Make and Manipulate Matrices
1.5 hours | Languages
Create and manipulate array data effectively using concatenation, creation functions, reshaping, and indexing.

Calculations with Vectors and Matrices
1 hour | Languages
Apply mathematical and statistical operations to vectors and matrices.

Matlab


You can write a long script and execute.
Script is saved as .m file.
You can execute a command here:
• Create or operate the variables
• Plot figure
• etc
Variables are stored here.

Some tips and basic commands

→ Matlab is an interpreted language without compiling.
→ When you use it to run a script, it executes line by line. It will **NOT** tell you an error somewhere in your script (such as a typo) until it executes that line.
→ Matlab has many ready functions and toolboxes, by the company or by individual. When you download and install Matlab, remember to select the **Signal Processing** toolbox.
→ Variables in the workspace should be saved manually. Otherwise, they will be deleted after Matlab is close.

→ Good habit: place ‘**;**’ at end of a statement to suppresses display of value (try to type ‘a = 0’ and ‘a = 0;’)

Basic commands

%	used to denote a comment
...	continues the statement on next line when a command is too long.

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To find out how to use a function, use command ‘**help**’.
For example, **help disp**

Matlab’s Workspace

save – save workspace vars to *.mat file.
load – load variables from *.mat file.
clear all – clear workspace vars.
close all – close all figures
clc – clear screen
clf – clear figure

Numbers

By default, MATLAB stores all numeric values as double-precision floating point.

Mathematical functions: **sqrt(x)**, **exp(x)**, **cos(x)**, **sin(x)**, **tan(x)**, **sum(x)**, **log(x)**, etc.
Operations: **+**, **-**, *, **/**
Power: **^**
Round toward negative infinity: **floor(x)**
Constants: **pi**, etc.

Generate random number: **rand(x)**

Absolute: **abs(x)**

Arrays and Matrices

v = [-2 3 0 4.5 -1.5]; % length 5 row vector.
v = v'; % transposes v.
v(1); % first element of v.
v(2:4); % entries 2-4 of v.
v([3,5]); % returns entries 3 & 5.
v=[4:-1:2]; % same as v=[4 3 2];

Example:
a=1:3; b=2:4;
c=[a b]; → c = [1 2 3 2 3 4];
c=[a; b]; → c = [1 2 3;
2 3 4];

Arrays and Matrices (2)

x = linspace(-pi,pi,10); % creates 10 linearly-spaced elements from –pi to pi.

Example:
A = [1 2 3; 4 5 6]; % creates 2x3 matrix
A(1,2) % the element in row 1, column 2.
A(:,2) % the second column.
A(2,:) % the second row.

Arrays and Matrices (3)

A+B, **A-B**, **2*A**, **A*B**	% matrix addition, matrix subtraction, scalar multiplication, matrix multiplication
A.*B	% element-by-element multiply.
A’	% transpose of A
det(A)	% determinant of A
cross(A, B)	% cross product

Creating special matrices

diag(v)	% change a vector v to a diagonal matrix.
diag(A)	% get diagonal of A.
eye(n)	% identity matrix of size n.
zeros(m,n)	% m-by-n zero matrix.
ones(m,n)	% m*n matrix with all ones.

Logical Conditions

== (equal)

~= (not equal)

<, **>**, **<=**, **>=** (greater or less)

~ (not)

& (element-wise logical and)

| (element-wise logical or)

Matrix/vector operations

length(v)	% determine length of vector.
size(A)	% determine size of matrix.
norm(A), norm(v)	% determine norm of matrix or vector.
fliplr(v)	% flip array left to right

For loops

```
x = 0;
for i=1:2:5       % start at 1, increment by 2
    x = x+i;        % end with 5.
end
```

This computes x = 0+1+3+5=9.

While loops

```
x=7;
while (x >= 0)
    x = x-2;
end;
```

This computes x = 7-2-2-2-2 = -1.

break – terminates execution of for and while loops. For nested loops, it exits the innermost loop only.

If statements

```
if (x == 3)
    disp(‘The value of x is 3.’);
elseif (x == 5)
    disp(‘The value of x is 5.’);
else
    disp(‘The value of x is not 3 or 5.’);
end;
```

Switch statement

```
switch face
case {1}
      disp('Rolled a 1');
case {2}
      disp('Rolled a 2');
otherwise
      disp('Rolled a number >= 3');
end
```

<u>NOTE:</u> Unlike C, ONLY the SWITCH statement between the matching case and the next case, otherwise, or end are executed. *(So breaks are unnecessary.)*

Vpa(simplify)

→ syms
→ Vpa()
→ Simplify()

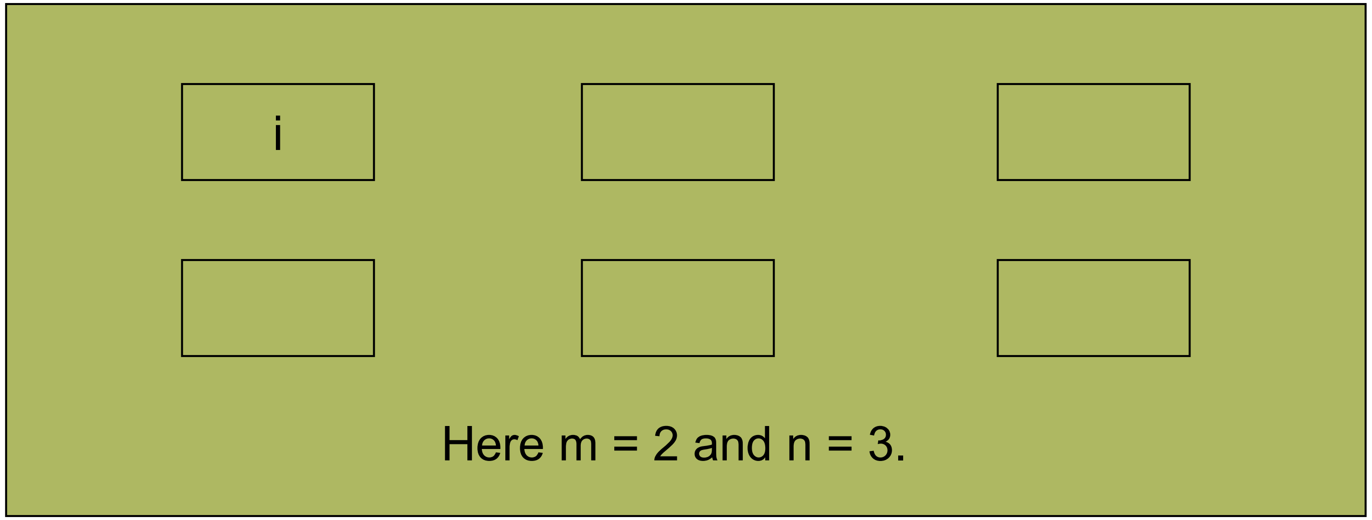
Graphics

```
x = linspace(-1,1,10);
y = sin(x);
plot(x,y);                    % plots y vs. x.
plot(x,y,'k-');               % plots a black line of y vs. x.

hold on;                      % put several plots in the same figure window.
figure(x);                    % open new figure window.
```

Graphics (2)

subplot(m,n,i) % Makes an m*n array for plots. Will place plot in the i^{th} position.



Example:

```
% Generate and plot a sine wave
fs = 1000; % Sampling frequency
t = 0:1/fs:1; % Time vector
f = 5; % Frequency (Hz)
x1 = sin(2*pi*f*t);
x2 = cos(2*pi*f*t);

subplot(2,1,1)
plot(t, x1);
xlabel('Time (s)');
ylabel('Amplitude');
title('Sine Wave');

subplot(2,1,2)
plot(t, x2,'k');
hold on; % Try to remove this line
plot([0,1], [0, 0], 'r--')
xlabel('Time (s)');
ylabel('Amplitude');
title('Cosine Wave');
```

Graphics (3)

plot3(x,y,z) % plot 2D function.
mesh(x,y_ax,z_mat) – surface plot.

axis([xmin xmax ymin ymax]) – change axes
title('My title'); - add title to figure;
xlabel, ylabel – label axes.
legend – add key to figure.

Exercise 1

→ Create a vector v which = [2,4,6,8,10] (try to use a ‘for’ loop)
→ Create a matrix m which = [0, 1, 2, 3, 4; 5, 6, 7, 8, 9; 0, 0.1, 0.2, 0.3, 0.4; 0.5, 0.6, 0.7, 0.8, 0.9]
→ Define tv which is the transpose of v
→ Calculate dot multiply res which m*tv
→ Plot res using * dots and in red color

Exercise 2

Create and plot an exponential sequence
$x(n) = 0.9^n,\ n \geq 0$

Decompose the sequence to a complex exponential
$x(n) = e^{j\omega_0 n} = \cos(\omega_0 n) + j\,\sin(\omega_0 n)$

Plot the real part (cosine)
And the imagine part (sine)

3D plot of the complex exponential